# Credit Risk Portfolio Analytics — Assignment Deliverables

**Data:** 5 provided tables (customers, loan_accounts, bureau_credit, repayment_history, collections_events) — 10,000 customers **Time:** 2-3 days | **Format:** Code repo/notebook + dashboard + short deck

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## 1. Data Ingestion

- Load and join all 5 tables (document your join logic / keys used)
- Handle missing values, inconsistent labels, and other data quality issues — state how, not just "dropped nulls"
- Output: one clean, analysis-ready dataset (+ a short data quality notes section)

## 2. Data Analytics

- Portfolio profiling across key dimensions: income, employment type, region, age, loan type
- 4-5 concrete insights with a "so what" (e.g., segment X = Y% of exposure, Z% higher default rate)
- At least one cross-dimensional cut (e.g., income × employment, or region × loan type)

## 3. Model Building

- Predict `target_default` using engineered features (aggregate bureau + repayment history — not just raw application fields)
- Baseline (logistic regression) + one ML model (XGBoost/LightGBM)
- Report AUC, KS, Gini + feature importance / SHAP
- One paragraph: how you'd monitor this model post-deployment (PSI/CSI, retraining cadence)

## 4. Dashboard

- Interactive (Streamliy/JSX/HTML – your choisce)
- **Portfolio view**: risk segments, exposure, trends across dimensions
- **Account-level view**: individual risk score + key drivers
- Minimum one working filter/drill-down

## 5. AI Feature (must be embedded in the dashboard, not standalone)

Pick **one**:

- **Anomaly detection** — flag accounts/customers with unusual repayment or bureau patterns (e.g., sudden DPD spike, outlier transaction behavior) vs. their own history or peer segment
- **"Ask the Portfolio" chat** — natural-language Q&A over the dashboard data (e.g., "which segment has the highest default rate this quarter?") using an LLM, grounded in the actual underlying data (no hallucinated numbers)
- State clearly: what triggers the AI feature, what a false positive/negative looks like, and the human-in-the-loop step

## Also Required

- **1-page product brief** — problem, target user, scope (in/out)
- **Architecture diagram** — ingestion → analytics → model → dashboard (+ AI feature), tools at each stage
- **10-min pitch deck** — presented as if to a bank's Chief Risk Officer; must include risk/compliance angle
- **Assumptions log** — anything invented or approximated, and what he'd need from a real bank's IT/data team to ship this